1. INTRODUCTION

Flexible and wearable electronics require conductors that can bend, stretch, and retain conductivity, a requirement that rigid metal wires and foils cannot fulfill effectively when integrated into biological interfaces. Conducting fibers, fabricated via routes such as solution spinning, coating, and electrospinning, are being rapidly developed to meet this need. In biomedical applications, particularly peripheral nerve regeneration, the specific challenge extends beyond basic conductivity to encompass fiber alignment. Moving from randomly oriented electrospun fiber mats to uniaxially aligned arrays provides directional conductivity and topographical cues that guide Schwann cell migration and axonal outgrowth. This report documents the experimental work completed thus far on the electrospinning of polycaprolactone (PCL) solutions to produce randomly oriented nanofibers, alongside the refined methodology for upcoming polymer blend trials (PLA/PCL) and solvent-variation trials using Trifluoroethanol (TFE) and Acetone mixtures.

2. LITERATURE REVIEW

2.1 Overview of Conducting Fibers

Conducting fibers are flexible polymer matrices engineered to carry electrical current, widely utilized in flexible energy devices and neural tissue engineering. Fabrication strategies typically include directly embedding conductive nanoparticles into the spinning solution or coating a conductive layer onto a pre-spun fiber. The central challenge in synthesizing these materials is achieving high electrical conductivity without compromising the polymer's inherent flexibility or biodegradability.

2.2 Work Done by Previous Researchers

Recent advancements have demonstrated the efficacy of incorporating conductive fillers into biodegradable polymers for nerve repair. Zhou et al. successfully embedded spherical polypyrrole (PPy) nanoparticles directly into electrospun poly(lactic acid) (PLA) nanofibers [1]. This embedded approach achieved stable conductivity and promoted strong Schwann cell attachment and elongation along aligned fibers [1]. Alternatively, Namhongsa et al. utilized a surface-coating strategy, combining 3D printing for structural support with electrospinning to create PLCL and PLGA scaffolds, which were subsequently surface-coated with PPy [2]. This dual-scale architecture enhanced both the conductivity and biocompatibility of the nerve guide conduits [2].

Carbon-based nanomaterials have also shown significant promise. Wang et al. developed a biomimetic, multichannel sponge using Antheraea pernyi silk fibroin (ApF),

PLCL, and Graphene Oxide (GO) [3]. The incorporation of GO improved the mechanical strength of the scaffold and promoted Schwann cell myelination to a degree comparable to clinical autografts [3]. Similarly, Jaswal et al. demonstrated that electrospinning PCL with reduced graphene oxide-enfolded gold nanoparticles (RGO-AuNPs) led to significant neuronal outgrowth in PC12 and S42 cell models [4].

To achieve the necessary uniaxial alignment for nerve regeneration, Li, Wang, and Xia pioneered the gap-electrode collector method, where electrostatic forces stretch a charged fiber across a gap between two grounded conductive strips, yielding highly aligned arrays [5].

2.3 Identified Research Gaps

While both conducting polymers (like PPy) and carbon-based nanomaterials (like RGO) have proven effective in individual studies, few investigations systematically compare their efficacy within the exact same deliberately aligned nanofibrous system. Furthermore, optimizing the mechanical flexibility-strength trade-off through specific solvent and polymer blending strategies requires further targeted experimentation.

3. OBJECTIVES

1. Design and evaluate a biomimetic, conductive, and aligned nanofibrous scaffold tailored specifically for peripheral nerve regeneration, capable of supporting Schwann cell guidance and axonal outgrowth.

2. Establish a reliable electrospinning process for PCL-based and PLA/PCL-based nanofibrous matrices.

3. Evaluate pure PCL electrospinning using varying ratios of a TFE/Acetone solvent mixture (20:80, 40:60, and 60:40) to leverage PCL's high solubility in TFE and eliminate needle-tip freezing issues.

4. Evaluate PLA:PCL blending ratios in acetone, specifically targeting 70:30 (for balanced flexibility), 80:20 (for superior tensile strength), and 50:50 (for high elasticity), to optimize overall mechanical properties.

5. Compare the efficacy of two distinct conductive fillers (Polypyrrole and RGO-AuNPs) to determine which yields the optimal conductivity profile for cellular stimulation.

6. Transition from randomly oriented fibers to uniaxially aligned arrays utilizing a rotating drum collector initially, with the gap-electrode collector as a secondary alternative.

4. PLAN OF WORK

The research plan is structured into three progressive phases: initial process establishment, polymer and solvent optimization, and final functionalization with conductive and topographical cues. The first phase, which has been successfully completed, involved a comprehensive literature review to identify current research gaps in conducting fibers and nerve guide conduits. Following this, fundamental electrospinning parameters were established in the laboratory. Baseline trials were conducted by electrospinning pure PCL dissolved in acetone at three distinct concentrations (13, 15, and 17 wt%) using a standard static flat-plate collector. This phase successfully produced random nanofiber mats but highlighted specific processing challenges, notably rapid solvent evaporation leading to solution freezing at the needle tip, which required further troubleshooting.

To address these challenges, the second phase of the project will focus on solvent variation and polymer blending. To resolve the needle-tip freezing issue, the project will pivot to electrospinning pure PCL utilizing a binary solvent system of 2,2,2-Trifluoroethanol (TFE) and Acetone. By testing solvent ratios of 20:80, 40:60, and 60:40 (TFE:Acetone), we aim to depress the evaporation rate and achieve a stable Taylor cone. In parallel, to optimize the mechanical strength and flexibility required for a peripheral nerve scaffold, blends of Poly(lactic acid) (PLA) and PCL will be electrospun in an acetone solvent. Ratios of 70:30, 80:20, and 50:50 (PLA:PCL) will be systematically evaluated to map out the mechanical property spectrum, balancing the stiffness of PLA with the high elasticity of PCL.

The final phase will integrate electrical conductivity and topographical alignment into the optimized polymer matrices. Once the ideal solvent and blend parameters are established, the project will incorporate and quantitatively compare two distinct conductive fillers: Polypyrrole (PPy) and RGO-AuNPs. Following the selection of the optimal conductive filler, the collection method will be upgraded from a static flat plate to a high-speed rotating drum collector to forcefully align the fibers along a single axis. The final composite scaffolds will then undergo rigorous characterization, including Scanning Electron Microscopy (SEM) for morphological alignment verification, four-point probe testing for electrical conductivity, and tensile testing to confirm suitability for peripheral nerve repair.

5. MATERIALS AND METHODS

5.1 Work Completed: Pure PCL Trials

Initial baseline trials utilized pure Polycaprolactone (PCL) dissolved in acetone to establish standard electrospinning parameters. Solutions were prepared at three distinct concentrations targeting a 5 g total mass per batch: 13 wt% (0.65 g PCL / 4.35 g Ace-

tone), 15 wt% (0.75 g PCL / 4.25 g Acetone), and 17 wt% (0.85 g PCL / 4.15 g Acetone). The preparation protocol involved thoroughly washing glass vials with soap and distilled water, followed by oven-drying at 80°C for 3 hours to ensure complete moisture removal. The polymer and solvent were then combined and subjected to continuous magnetic stirring for 24 hours to promote homogenization. Electrospinning was subsequently conducted under ambient conditions (approximately 25°C). The operational parameters were maintained at an applied voltage of 2-5 kV, a controlled solution flow rate of 1-2 mL/hr, and a fixed tip-to-collector distance of 15 cm, utilizing a grounded static flat-plate collector to gather the resulting random nanofiber mats.

6. RESULTS AND DISCUSSION

Baseline electrospinning was successfully executed at all three pure PCL concentrations (13%, 15%, and 17%), yielding randomly oriented nanofiber mats. During these initial trials, two primary processing issues were documented. First, inadequate solution homogeneity occasionally disrupted the Taylor cone, indicating that more vigorous or prolonged magnetic stirring is required prior to spinning. Second, rapid solvent evaporation caused the polymer solution to freeze at the needle tip. This issue directly motivated the planned transition to the TFE/Acetone binary solvent system (20:80, 40:60, 60:40 ratios), which is expected to sufficiently depress the evaporation rate and stabilize the jet.

7. CONCLUSION

The development of a conductive, aligned nanofibrous scaffold requires carefully balancing electrical conductivity, mechanical flexibility, and morphological stability. Baseline trials successfully established standard PCL electrospinning protocols while revealing specific solvent-evaporation limitations. The upcoming experimental phase will directly address these limitations by transitioning to a controlled TFE/Acetone binary solvent system (at 20:80, 40:60, and 60:40 ratios) and systematically evaluating PLA/PCL blends (70:30, 80:20, 50:50) to optimize mechanical properties. Ultimately, these optimized matrices will host either PPy or RGO-AuNP fillers and be spun onto a rotating drum collector to achieve the critical uniaxial alignment required for effective peripheral nerve regeneration.

REFERENCES

[1] Zhou, J.F., Wang, Y.G., Cheng, L., Wu, Z., Sun, X.D., Peng, J., Preparation of polypyrrole-embedded electrospun poly(lactic acid) nanofibrous scaffolds for nerve tissue engineering. *Neural Regeneration Research*, 2016, 11, 1644-1652.

[2] Namhongsa, M., Daranarong, D., Sriyai, M., Molloy, R., Ross, S., Ross, G.M., Tuantranont, A., Tocharus, J., Sivasinprasasn, S., Topham, P.D., Tighe, B., Punyodom, W., Surface-Modified Polypyrrole-Coated PLCL and PLGA Nerve Guide Conduits Fabricated by 3D Printing and Electrospinning. *Biomacromolecules*, 2022, 23, 4532-4546.

[3] Wang, J., Cheng, Y., Wang, H., Wang, Y., Zhang, K., Fan, C., Wang, H., Mo, X., Biomimetic and hierarchical nerve conduits from multifunctional nanofibers for guided peripheral nerve regeneration. *Acta Biomaterialia*, 2020, 117, 180-191.

[4] Jaswal, et al., Nanographene enfolded AuNPs sophisticatedly synchronized polycaprolactone-based electrospun nanofibre scaffold for peripheral nerve regeneration. *Materials Science and Engineering: C*, 2020, 116, 111213.

[5] Li, D., Wang, Y., Xia, Y., Electrospinning of Uniaxially Aligned Nanofiber Arrays. *Nano Letters*, 2003, 3, 1167-1171.